1.2 Solutions — Real analysis I

Solutions to problems.md. Where a proof is a standard theorem, the version here is the one you should be able to reproduce closed-book.

Warm-up

1.

$1 - \frac{1}{n} < 1$ for every n, so 1 is an upper bound. Let $\varepsilon > 0$. By the Archimedean property there is n with $\frac{1}{n} < \varepsilon$, and then $1 - \frac{1}{n} > 1 - \varepsilon$. By the ε-characterisation of the supremum, $\sup = 1$. It is **not attained**, since $1 - \frac{1}{n} \neq 1$ for every n.

2.

Scratch work.

$$\left|\frac{n^2+1}{2n^2-3} - \frac{1}{2}\right| = \frac{|2n^2+2-2n^2+3|}{2(2n^2-3)} = \frac{5}{2(2n^2-3)}.$$

For $n \geq 2$ we have $n^2 \geq 3$, so $2n^2 - 3 \geq n^2 > 0$, and the expression is at most $\frac{5}{2n^2} \leq \frac{5}{2n}$.

Proof. Let $\varepsilon > 0$ and choose an integer $N \geq 2$ with $N > \frac{5}{2\varepsilon}$. For $n \geq N$,

$$\left|\frac{n^2+1}{2n^2-3} - \frac{1}{2}\right| \leq \frac{5}{2n} \leq \frac{5}{2N} < \varepsilon. \qquad \square$$

3.

(a) Suppose $a_n \to a$ and $a_n \to b$ with $a \neq b$. Let $\varepsilon = |a-b|/2 > 0$. For n beyond both thresholds, $|a-b| \leq |a-a_n| + |a_n - b| < 2\varepsilon = |a-b|$, a contradiction.

(b) With $\varepsilon = 1$ there is N with $|a_n - a| < 1$, and hence $|a_n| < |a| + 1$, for $n \geq N$. So $|a_n| \leq \max\{|a_1|, \dots, |a_{N-1}|, |a|+1\}$ for all n.

4.

(a) **Converges.** Ratio test:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n} \to \frac{1}{e} < 1.$$

(b) **Diverges.** First, $n^{1/n} \to 1$: write $n^{1/n} = 1 + h_n$ with $h_n \geq 0$. By the binomial theorem, $n = (1+h_n)^n \geq \binom{n}{2} h_n^2$, so $0 \leq h_n \leq \sqrt{2/(n-1)} \to 0$ for $n \geq 2$. Hence

$$\frac{1/n^{1+1/n}}{1/n} = \frac{1}{n^{1/n}} \to 1,$$

and by limit comparison with the divergent harmonic series, the series diverges.

(c) **Diverges.** $|a_n| = \frac{n}{n+1} \to 1$, so $a_n \not\to 0$, and the nth-term test applies.

(d) **Converges, with sum 1.** The series telescopes: the partial sum is $s_N = 1 - \frac{1}{\sqrt{N+1}} \to 1$.

5.

Continuity at 0: $|f(x) - f(0)| = |x|\,|\sin(1/x)| \le |x|$. Given ε, take $\delta = \varepsilon$.

Not differentiable at 0: the difference quotient is $\dfrac{f(h) - f(0)}{h} = \sin(1/h)$. Along $h_n = \frac{1}{2\pi n} \to 0$ it equals 0. Along $k_n = \frac{1}{2\pi n + \pi/2} \to 0$ it equals 1. By the sequential characterisation, the limit as $h \to 0$ does not exist.

6.

Claim: $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for $x, y \ge 0$.

Assume $x \ge y$. Then $\left(\sqrt{y} + \sqrt{x-y}\right)^2 = x + 2\sqrt{y(x-y)} \ge x$, so $\sqrt{x} \le \sqrt{y} + \sqrt{x-y}$, which is the claim.

Given $\varepsilon > 0$, let $\delta = \varepsilon^2$. Then $|x-y| < \delta$ implies $|\sqrt{x}-\sqrt{y}| < \varepsilon$. So $\sqrt{x}$ is **uniformly continuous**.

Not Lipschitz: $\dfrac{|\sqrt{x} - \sqrt{0}|}{|x - 0|} = \dfrac{1}{\sqrt{x}} \to \infty$ as $x \to 0^+$, so no constant M works.

Core

7.

(a) Let (a_n) be increasing and bounded, and let $s = \sup\{a_n\}$, which exists by completeness. Given $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound, so $a_N > s - \varepsilon$ for some N. For $n \ge N$, $s - \varepsilon < a_N \le a_n \le s$, so $|a_n - s| < \varepsilon$. The decreasing case is symmetric, with inf.

(b) *Bounded above by 2* (induction): $a_1 = \sqrt{2} < 2$, and if $a_n < 2$ then $a_{n+1} = \sqrt{2 + a_n} < \sqrt{4} = 2$. Also $a_n > 0$.

Increasing: $a_{n+1}^2 - a_n^2 = 2 + a_n - a_n^2 = (2 - a_n)(1 + a_n) > 0$. Since both terms are positive, $a_{n+1} > a_n$.

By MCT, $a_n \to L$ for some L with $\sqrt{2} \le L \le 2$. Squaring the recursion gives $a_{n+1}^2 = 2 + a_n$. Letting $n \to \infty$ (limit laws), $L^2 = 2 + L$, so $L = 2$ or $L = -1$. Since $L > 0$, $L = 2$.

8.

Lemma. Every sequence has a monotone subsequence.

Call m a *peak* if $a_m \ge a_n$ for all $n > m$.

- If there are infinitely many peaks $m_1 < m_2 < \cdots$, then $a_{m_1} \ge a_{m_2} \ge \cdots$, a decreasing subsequence.
- If there are finitely many, let n_1 be larger than all of them. Since n_1 is not a peak, there is $n_2 > n_1$ with $a_{n_2} > a_{n_1}$. Since n_2 is not a peak, there is $n_3 > n_2$ with $a_{n_3} > a_{n_2}$. Continuing (by induction) gives an increasing subsequence.

Bolzano–Weierstrass. If (a_n) is bounded, its monotone subsequence is bounded, so it converges by MCT. □

9.

(a) Let (a_n) be Cauchy.

Bounded: with $\varepsilon = 1$, choose N with $|a_n - a_N| < 1$ for $n \geq N$. Then $|a_n| \leq \max\{|a_1|, \dots, |a_{N-1}|, |a_N| + 1\}$.

Subsequence: by Bolzano–Weierstrass, $a_{n_k} \to a$ for some a.

Whole sequence: given $\varepsilon > 0$, choose N with $|a_m - a_n| < \varepsilon/2$ for $m, n \geq N$, then k with $n_k \geq N$ and $|a_{n_k} - a| < \varepsilon/2$. For $n \geq N$, $|a_n - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a| < \varepsilon$.

(b) $H_{2n} - H_n = \sum_{k=n+1}^{2n} \frac{1}{k} \geq n \cdot \frac{1}{2n} = \frac{1}{2}$, since each of the n terms is at least $\frac{1}{2n}$. So with $\varepsilon = \frac{1}{2}$, for every N the indices $m = N$ and $n = 2N$ give $|H_n - H_m| \geq \frac{1}{2}$. The sequence is not Cauchy, hence it diverges, so $\sum 1/k$ diverges.

10.

(a) Let $s_n = \sum_{j=1}^{n} a_j$ and $t_k = \sum_{j=0}^{k} 2^j a_{2^j}$. Both sequences increase, so each converges iff it is bounded (MCT).

Upper bound. Group $s_{2^{k+1}-1}$ into blocks of length $1, 2, 4, \dots, 2^k$, and bound each term in a block by the block's first term (since a_n is decreasing):

$$s_{2^{k+1}-1} = a_1 + (a_2 + a_3) + (a_4 + \cdots + a_7) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}-1}) \leq a_1 + 2a_2 + 4a_4 + \cdots + 2^k a_{2^k} = t_k.$$

So if (t_k) is bounded, so is (s_n) (every n is at most some $2^{k+1} - 1$).

Lower bound. Group s_{2^k} as $a_1 + a_2 + (a_3 + a_4) + (a_5 + \cdots + a_8) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k})$, and bound each term in a block below by the block's last term:

$$s_{2^k} \geq a_1 + a_2 + 2a_4 + 4a_8 + \cdots + 2^{k-1} a_{2^k} \geq \frac{1}{2}\left(a_1 + 2a_2 + 4a_4 + \cdots + 2^k a_{2^k}\right) = \frac{t_k}{2}.$$

So if (s_n) is bounded, so is (t_k). Hence the two series converge or diverge together. □

(b) If $p \leq 0$, then $1/n^p \not\to 0$ and the series diverges. If $p > 0$, then $a_n = n^{-p}$ is decreasing and positive, and

$$\sum_k 2^k a_{2^k} = \sum_k 2^k 2^{-kp} = \sum_k \left(2^{1-p}\right)^k,$$

a geometric series that converges iff $2^{1-p} < 1$, that is, iff $p > 1$.

11.

($\Leftarrow$) Let K be closed and bounded and (x_n) a sequence in K. By Bolzano–Weierstrass, $x_{n_k} \to x$ for some $x \in \mathbb{R}$. If $x_{n_k} = x$ for infinitely many k, then $x \in K$. Otherwise x is a limit point of K, and since K is closed, $x \in K$.

($\Rightarrow$) *Bounded:* if not, choose $x_n \in K$ with $|x_n| > n$. Every subsequence is unbounded, so none converges. This contradicts sequential compactness.

Closed: let x be a limit point of K and choose $x_n \in K$ with $x_n \to x$. By hypothesis some subsequence converges to a point of K. But every subsequence converges to x, so $x \in K$.

12.

(a) Let K be compact, f continuous on K, and (y_n) a sequence in $f(K)$, say $y_n = f(x_n)$ with $x_n \in K$. By compactness, $x_{n_k} \to x \in K$. By (sequential) continuity, $y_{n_k} = f(x_{n_k}) \to f(x) \in f(K)$. So $f(K)$ is sequentially compact.

(b) $f([a,b])$ is compact, hence closed and bounded (problem 11), and non-empty. So $M = \sup f([a,b])$ exists. Since $f([a,b])$ is closed, it contains M: either M is in the set, or M is a limit point of it. Hence $M = f(c)$ for some c. The minimum is similar.

(c) On $(0,1)$, $f(x) = x$ is continuous with no maximum or minimum, and $f(x) = 1/x$ is unbounded. On $[0,1]$, the discontinuous function $f(x) = x$ for $x < 1$, $f(1) = 0$, has supremum 1, which is not attained.

13.

Let $S = \{x \in [a,b] : f(x) < L\}$. S is non-empty ($a \in S$) and bounded above by b. Let $c = \sup S$; then $a \le c \le b$.

Suppose $f(c) < L$. Then $c \neq b$ (since $f(b) > L$). By continuity with $\varepsilon = L - f(c)$, there is $\delta > 0$ such that $f(x) < L$ for $x \in [a,b]$ with $|x - c| < \delta$. Shrinking δ, we may assume $c + \delta/2 \le b$. Then $c + \delta/2 \in S$, contradicting that c is an upper bound.

Suppose $f(c) > L$. Then $c \neq a$. By continuity there is $\delta \in (0, c-a)$ with $f(x) > L$ for $x \in (c-\delta, c]$. No point of S lies in $(c-\delta, c]$, and no point of S exceeds c, so $c - \delta$ is an upper bound for S smaller than c. This is a contradiction.

Hence $f(c) = L$, and $c \neq a, b$ since $f(a) < L < f(b)$. □

14.

Suppose f is continuous on $[a,b]$ but not uniformly continuous. Then there are $\varepsilon_0 > 0$ and points $x_n, y_n \in [a,b]$ with $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \ge \varepsilon_0$ (negate the definition with $\delta = 1/n$). By Bolzano–Weierstrass, $x_{n_k} \to x$, and $x \in [a,b]$ since the interval is closed. Then $y_{n_k} \to x$ too, because $|y_{n_k} - x| \le |y_{n_k} - x_{n_k}| + |x_{n_k} - x|$. By continuity at x, $f(x_{n_k}) - f(y_{n_k}) \to f(x) - f(x) = 0$, contradicting $|f(x_{n_k}) - f(y_{n_k})| \ge \varepsilon_0$. □

$1/x$ on $(0,1)$: it is continuous as a quotient of continuous functions with a non-zero denominator. Take $x_n = \frac{1}{n}$ and $y_n = \frac{1}{n+1}$ (for $n \ge 2$, both in $(0,1)$). Then $|x_n - y_n| \to 0$, but $|f(x_n) - f(y_n)| = 1$. By the sequential criterion, $1/x$ is not uniformly continuous.

15.

(a) **Rolle.** If f is continuous on $[a,b]$, differentiable on (a,b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a,b)$.

Proof. By EVT, f attains a maximum M and a minimum m on $[a,b]$. If both are attained only at endpoints, then $M = m = f(a)$ and f is constant, so $f' = 0$ everywhere. Otherwise an extremum is attained at an interior point c.

At an interior maximum c, for $h > 0$ we have $\frac{f(c+h) - f(c)}{h} \le 0$, and for $h < 0$ we have $\frac{f(c+h) - f(c)}{h} \ge 0$. Taking limits, $f'(c) \le 0$ and $f'(c) \ge 0$, so $f'(c) = 0$. The minimum is similar. □

MVT. Let $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$. Then $g(a) = g(b) = f(a)$, and Rolle gives c with $0 = g'(c) = f'(c) - \frac{f(b)-f(a)}{b-a}$.

(b) For $a < x < y < b$, the MVT gives $f(y) - f(x) = f'(c)(y - x) > 0$.

(c) Let $g(x) = x - \ln(1+x)$ for $x \geq 0$. Then $g(0) = 0$ and $g'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} > 0$ for $x > 0$. For $x > 0$, the MVT on $[0, x]$ gives $g(x) = g(x) - g(0) = g'(c)\, x > 0$ for some $c \in (0, x)$.

16.

(a) Taylor's theorem for e^x at $x_0 = 0$, evaluated at $x = 1$:

$$e = \sum_{k=0}^{n} \frac{1}{k!} + \frac{e^c}{(n+1)!}, \quad 0 < c < 1.$$

The error is $\frac{e^c}{(n+1)!} < \frac{3}{(n+1)!}$. We need $(n+1)! > 3000$. Since $6! = 720$ and $7! = 5040$, take $n = 6$: $\sum_{k=0}^{6} \frac{1}{k!} = 2.71805\ldots$, with error less than $3/5040 < 6 \times 10^{-4}$. (The true error is about 2.3×10^{-4}.)

(b) Taylor's theorem for cos at 0 with $n = 1$:

$$\cos x = \cos 0 - (\sin 0)\, x + \frac{-\cos c}{2} x^2 = 1 - \frac{\cos c}{2} x^2$$

for some c between 0 and x (for $x = 0$ the claim is trivial). Since $\cos c \leq 1$, $\cos x \geq 1 - \frac{x^2}{2}$.

17.

Let $P_n = \{0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\}$. Since x^2 is increasing on $[0, 1]$, on $[\frac{k-1}{n}, \frac{k}{n}]$ the sup is $\frac{k^2}{n^2}$ and the inf is $\frac{(k-1)^2}{n^2}$. So

$$U(f, P_n) = \sum_{k=1}^{n} \frac{k^2}{n^3} = \frac{(n+1)(2n+1)}{6n^2}, \qquad L(f, P_n) = \sum_{k=1}^{n} \frac{(k-1)^2}{n^3} = \sum_{j=0}^{n-1} \frac{j^2}{n^3} = \frac{(n-1)(2n-1)}{6n^2}.$$

Then

$$U(f, P_n) - L(f, P_n) = \frac{(2n^2 + 3n + 1) - (2n^2 - 3n + 1)}{6n^2} = \frac{1}{n} \to 0,$$

so f is integrable by the criterion. Both $U(f, P_n)$ and $L(f, P_n)$ tend to $\frac{2}{6} = \frac{1}{3}$. Since $L(f, P_n) \leq \int_0^1 f \leq U(f, P_n)$ for every n, $\int_0^1 x^2\, dx = \frac{1}{3}$.

18.

(a) Let $\varepsilon > 0$. By uniform convergence choose N with $|f_N(x) - f(x)| < \varepsilon/3$ for **all** $x \in A$. By continuity of f_N at c choose δ with $|f_N(x) - f_N(c)| < \varepsilon/3$ when $|x - c| < \delta$. Then for such x,

$$|f(x) - f(c)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| < \varepsilon.$$

(b) $\left| \int_a^b f_n - \int_a^b f \right| \leq \int_a^b |f_n - f| \leq (b - a) \sup_{[a,b]} |f_n - f| \to 0.$

(The hypothesis that f is integrable is in fact automatic; see notes.md §8.2.)

19.

Let $M = \sup_{[a,b]} |f|$, which is finite because integrable functions are bounded. For $x < y$ in $[a, b]$,

$$|G(y) - G(x)| = \left| \int_x^y f \right| \leq M(y - x),$$

so G is Lipschitz, hence continuous.

Let f be continuous at $c \in (a, b)$, and let $\varepsilon > 0$. Choose δ such that $|f(t) - f(c)| < \varepsilon$ for $|t - c| < \delta$. For $0 < |h| < \delta$ (with $c + h \in [a, b]$), using $\int_c^{c+h} f(c)\, dt = hf(c)$,

$$\left| \frac{G(c + h) - G(c)}{h} - f(c) \right| = \frac{1}{|h|} \left| \int_c^{c+h} (f(t) - f(c))\, dt \right| \leq \frac{1}{|h|} \cdot |h|\, \varepsilon = \varepsilon.$$

Hence $G'(c) = f(c)$. □

20.

(a) $f_n(0) = 0$. For $x > 0$, $0 \leq f_n(x) \leq \frac{nx}{n^2x^2} = \frac{1}{nx} \to 0$. So the pointwise limit is $f = 0$.

(b) $f_n(1/n) = \frac{1}{1+1} = \frac{1}{2}$, so $\sup_{[0,1]} |f_n - 0| \geq \frac{1}{2}$ for all n, which does not tend to 0. The convergence is not uniform. (The "bump" moves towards 0 without shrinking.)

(c) On $[\delta, 1]$, $|f_n(x)| \leq \frac{1}{nx} \leq \frac{1}{n\delta} \to 0$, uniformly in x.

21.

$|f_n(x)| \leq \frac{1}{\sqrt{n}}$ for all x, so $\sup |f_n| \to 0$ and $f_n \to 0$ uniformly. The limit 0 has derivative 0.

But $f_n'(x) = \sqrt{n} \cos(nx)$, and at $x = 0$, $f_n'(0) = \sqrt{n} \to \infty$. So $f_n'(0) \not\to 0$.

Moral: uniform convergence of f_n does not let you differentiate the limit term by term. You need uniform convergence of the *derivatives* f_n' (plus convergence of f_n at one point). This is the differentiation theorem in notes.md §8.3.

Challenge

22.

(a) *At a rational $r = p/q$:* $t(r) = 1/q > 0$. Irrationals $x_n \to r$ exist (for example $r + \sqrt{2}/n$, intersected with $[0, 1]$ by choosing the sign suitably), and $t(x_n) = 0 \not\to t(r)$. So t is discontinuous at r.

At an irrational x: let $\varepsilon > 0$. The set $F_\varepsilon = \{y \in [0, 1] \cap \mathbb{Q} : t(y) \geq \varepsilon\}$ consists of the fractions p/q with $q \leq 1/\varepsilon$ and $0 \leq p \leq q$. It is finite, and it does not contain x. Let $\delta = \min_{y \in F_\varepsilon} |y - x| > 0$. If $|y - x| < \delta$, then $y \notin F_\varepsilon$, so $t(y) < \varepsilon$, that is, $|t(y) - t(x)| < \varepsilon$. So t is continuous at x.

(b) *Lower sums.* Every subinterval contains an irrational, where $t = 0$, and $t \geq 0$, so $L(t, P) = 0$ for every P.

Upper sums. Let $\varepsilon > 0$ and let $F = F_{\varepsilon/2} = \{r_1 < \cdots < r_m\}$, finite by (a). Choose $\eta > 0$ with $\eta < \frac{\varepsilon}{4m}$ and small enough that the intervals $[r_i - \eta, r_i + \eta]$ are pairwise disjoint. Let P consist of 0, 1 and the points $r_i \pm \eta$ that lie in $(0, 1)$. The subintervals of P are of two kinds:

- those that contain a point of F: these are contained in the intervals $[r_i - \eta, r_i + \eta]$, so their total length is at most $2\eta m < \varepsilon/2$, and $\sup t \le 1$ on each;
- the rest, on which $\sup t \le \varepsilon/2$, and whose total length is at most 1.

So $U(t, P) < \frac{\varepsilon}{2} \cdot 1 + 1 \cdot \frac{\varepsilon}{2} = \varepsilon$. Hence $U(t) = 0 = L(t)$, so t is integrable with $\int_0^1 t = 0$.

(This agrees with Lebesgue's criterion: the discontinuity set $\mathbb{Q} \cap [0, 1]$ is countable, so it has measure zero.)

23.

(a) **M-test.** If $|f_n(x)| \le M_n$ on A and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly on A.

Proof. Let $S_n = \sum_{k=1}^{n} f_k$. For $n > m$ and all $x \in A$,

$$|S_n(x) - S_m(x)| \le \sum_{k=m+1}^{n} M_k,$$

which is less than ε for $m \ge N$ by the Cauchy criterion for $\sum M_k$. So (S_n) is uniformly Cauchy. For each x, $S_n(x)$ converges to some $S(x)$ (Cauchy in $\mathbb{R}$). Letting $n \to \infty$ in the inequality gives $|S(x) - S_m(x)| \le \varepsilon$ for all x and $m \ge N$. So the convergence is uniform. □

(b) $\left|\frac{\sin(nx)}{n^3}\right| \le \frac{1}{n^3}$, and $\sum \frac{1}{n^3} < \infty$. So the series converges uniformly on $\mathbb{R}$. Its partial sums are continuous, so g is continuous (problem 18a).

The termwise derivatives are $\frac{\cos(nx)}{n^2}$. They are continuous, and $\left|\frac{\cos(nx)}{n^2}\right| \le \frac{1}{n^2}$, so $\sum \frac{\cos(nx)}{n^2}$ converges uniformly on $\mathbb{R}$. The original series converges at $x = 0$ (indeed everywhere). By the differentiation theorem for series (applied on each interval $[-A, A]$), g is differentiable and $g'(x) = \sum \frac{\cos(nx)}{n^2}$. Since this is again a uniform limit of continuous functions, g is in fact C^1.

(Differentiating once more would give $-\sum \frac{\sin(nx)}{n}$, for which the M-test fails. The argument stops here.)

24.

Suppose f is continuous with $f(f(x)) = -x$ for all x.

f is injective: if $f(x) = f(y)$ then $-x = f(f(x)) = f(f(y)) = -y$.

A continuous injective $f : \mathbb{R} \to \mathbb{R}$ is strictly monotone. Let $T = \{(x, y) \in \mathbb{R}^2 : x < y\}$ and $D(x, y) = f(y) - f(x)$. By injectivity, $D \neq 0$ on T. Suppose $D(p) > 0$ and $D(q) < 0$ for some $p, q \in T$. T is convex, so the segment $s \mapsto (1 - s)p + sq$, $s \in [0, 1]$, stays in T. The function $s \mapsto D((1-s)p+sq)$ is continuous on $[0, 1]$ and changes sign, so by the IVT it vanishes somewhere, a contradiction. Hence D has constant sign on T, so f is strictly increasing or strictly decreasing.

Conclusion. If f is strictly increasing, so is $f \circ f$. If f is strictly decreasing, then $f \circ f$ is strictly increasing (a composition of two decreasing functions). Either way $f \circ f$ is strictly increasing, but $x \mapsto -x$ is strictly decreasing. This is a contradiction, so no such f exists.

(Discontinuous solutions do exist. Continuity is essential.)

25.

Let $S = \sum_{n=1}^{\infty} a_n$, and let $t_m = \sum_{k=1}^{m} a_{\sigma(k)}$ be the partial sums of the rearrangement. Let $\varepsilon > 0$. Since $\sum |a_n|$ converges, choose N with $\sum_{n>N} |a_n| < \varepsilon/2$. Then also

$$\left| S - \sum_{n=1}^{N} a_n \right| = \left| \sum_{n>N} a_n \right| \leq \sum_{n>N} |a_n| < \frac{\varepsilon}{2}.$$

Let $M = \max\{\sigma^{-1}(1), \dots, \sigma^{-1}(N)\}$. Then for $m \geq M$, the indices $\sigma(1), \dots, \sigma(m)$ include all of $1, \dots, N$. So $t_m - \sum_{n=1}^{N} a_n$ is a sum of terms a_j with distinct indices $j > N$, and

$$\left| t_m - \sum_{n=1}^{N} a_n \right| \leq \sum_{j>N} |a_j| < \frac{\varepsilon}{2}.$$

By the triangle inequality, $|t_m - S| < \varepsilon$ for all $m \geq M$. So the rearrangement converges to S. $\square$

(The rearranged series also converges absolutely: apply the same argument to $\sum |a_n|$.)